LLM-Powered Intelligent Document Analyzer for Educational Content Understanding

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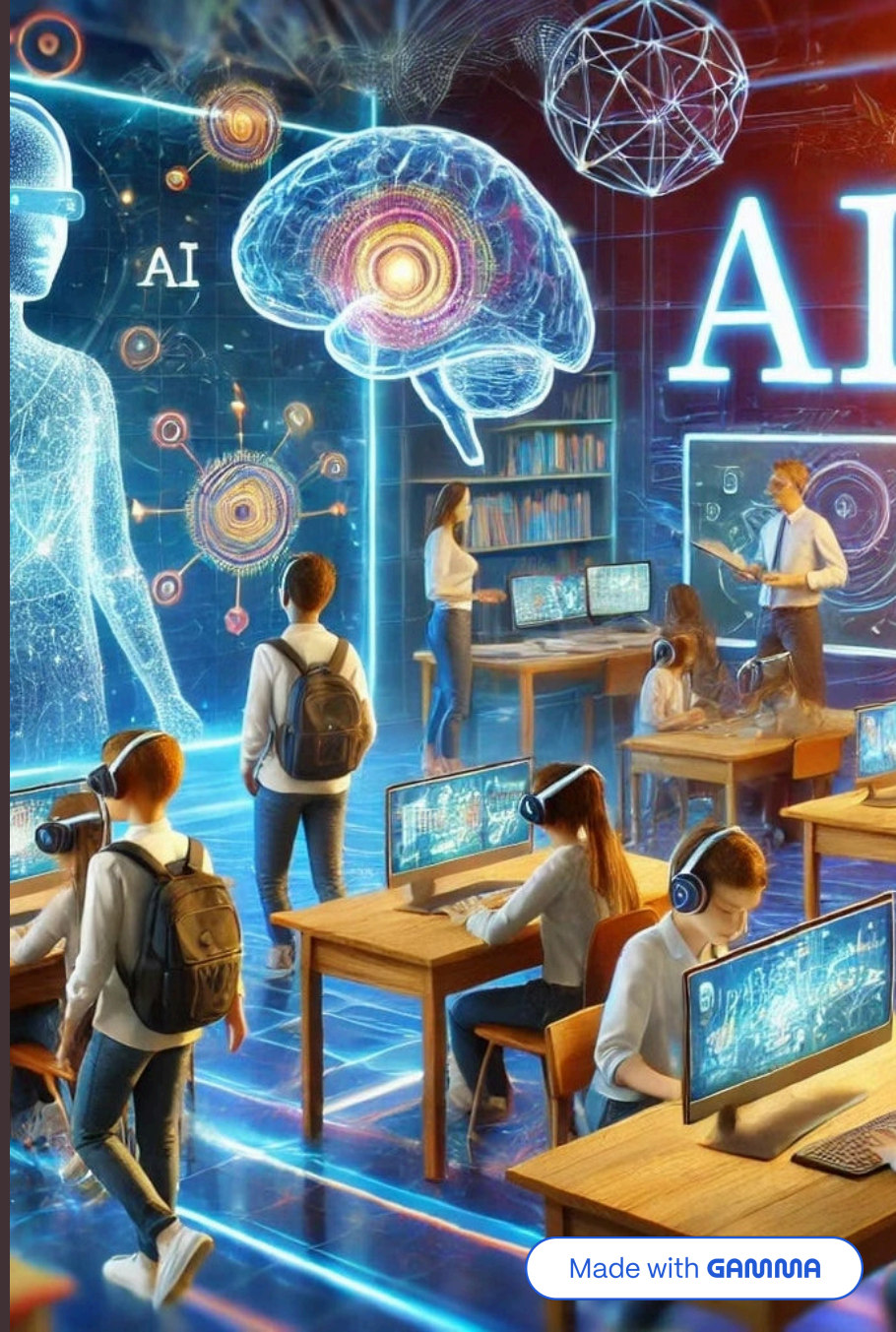

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Problem & Motivation

Content Explosion

Digital educational content is growing exponentially, creating information overload for educators and students alike.

Traditional NLP Limitations

Keyword extraction and shallow summarization fail to capture deep semantic meaning and educational intent.

Multimodal Blind Spot

Images, equations, diagrams, and visual elements are systematically ignored by text-only approaches.

Curriculum Gap

Lack of intelligent tools aligned with pedagogical frameworks and learning objectives.

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Proposed Solution

Intelligent Document Analysis

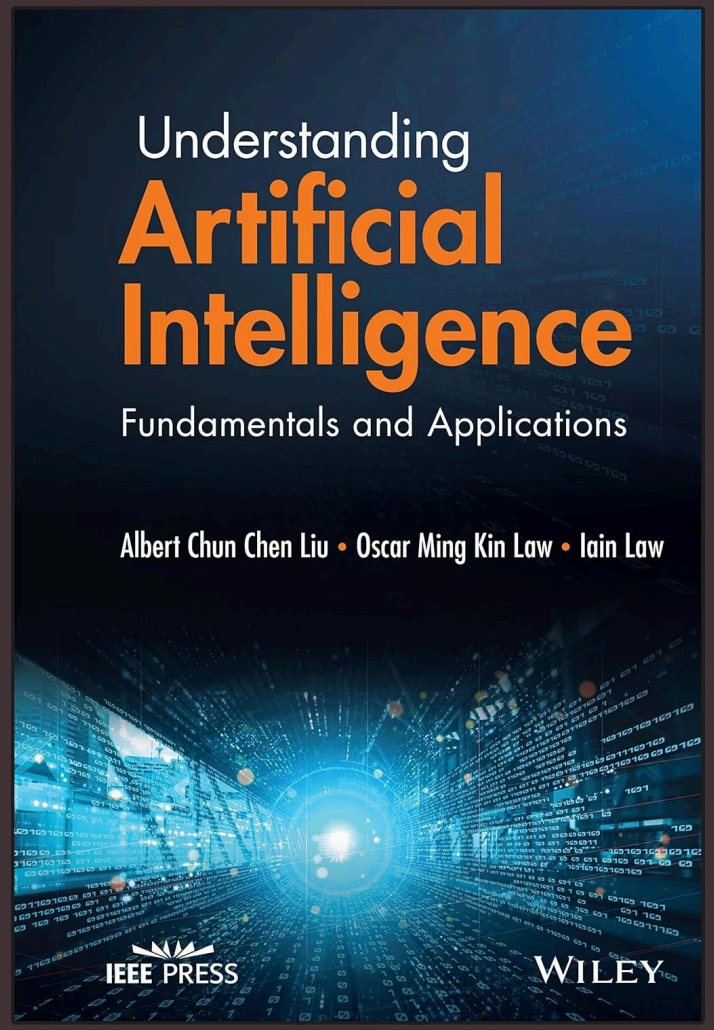
An LLM-assisted system powered by GPT-4 and GPT-4V that transforms educational content into actionable insights through multimodal understanding.

Core Capabilities

- Deep document summarization with semantic coherence
- Automatic concept identification and taxonomy mapping
- Context-aware question answering and generation
- Socratic questioning with chain-of-thought reasoning

Technical Foundation

Integrated OCR pipelines, vision models, knowledge graph construction, and curriculum-aligned ontologies create a comprehensive analysis framework.



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System Pipeline Architecture

Input PDF
Raw document ingestion

Text Extraction
OCR and layout analysis

Semantic Chunking
Context-aware segmentation

Cognitive Feature Extraction
Concept detection and tagging

Multi-LLM Ensemble
Coordinated reasoning

Bloom Concept Mapper
Learning taxonomy alignment

Entropy & Coherence Analyzer
Quality metrics computation

Web Dashboard
Interactive visualization

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Key System Modules

OCR & Preprocessing

Tesseract and Google Vision APIs extract text, while GPT-4V generates detailed image captions and equation interpretations for complete multimodal coverage.

Knowledge Graph Construction

Automatically builds curriculum-aligned ontologies by mapping detected concepts to standardized educational frameworks and prerequisite relationships.

Multi-Level Summarization

Generates section-level and global document summaries with varying detail levels, maintaining semantic coherence and pedagogical structure throughout.

Quiz Generation

Creates diverse question formats including fill-in-the-blank, multiple-choice, and Bloom's taxonomy-aligned items with validated distractors.

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Prompt Engineering Strategy

01

Instruction-Tuned Prompts

Task-specific templates optimized for each analysis function

02

Socratic Multi-Turn

Iterative questioning for deep concept extraction

03

Chain-of-Thought

Outline subtopics, then explain each systematically

04

Two-Step MCQ

Separate question+answer from distractor generation

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Novel Evaluation Metrics

Lexical Entropy

$$H(X) = -\sum_i p(x_i) \log p(x_i)$$

Measures vocabulary diversity and distribution across document sections.

Semantic Coherence

Average cosine similarity between adjacent sentence embeddings indicates logical flow quality.

Concept Recall

$$\text{Recall} = \frac{|\text{Shared Concepts}|}{|\text{Total Reference Concepts}|}$$

Identifies curriculum-aligned concepts detected versus expert-annotated reference set.

Error Rates

Hallucination rate and distractor error rate measure factual accuracy and distractor quality.

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Experimental Setup

MOOC Transcripts

Video lecture transcripts with timestamps and annotations

STEM Textbooks

Mathematics, physics, and computer science materials with equations

Student Reports

Research papers and project documentation

Baselines

TF-IDF pipeline and vanilla GPT-3.5

Evaluation Metrics

ROUGE-L, BERTScore, Concept Recall, Semantic Coherence

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Results & Bloom's Taxonomy Analysis

3%

Hallucination Rate

Down from 12% baseline

7%

Distractor Error

Reduced from 20%

Bloom's Taxonomy Heatmap Insights

Concept density analysis reveals dominance of lower-order skills (Remember, Understand) with underrepresentation of higher-order thinking (Analyze, Evaluate). This guides curriculum development toward critical thinking content.

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Future Work & Conclusion

Future Directions

- **Adaptive Personalization**

  Student knowledge gap detection and customized content recommendations

- **Multimodal Expansion**

  Video and audio processing with keyframe analysis and speech-to-text

- **Conversational Tutor**

  Interactive dialogue interface for real-time student support

- **Large-Scale Evaluation**

  Cross-subject and cross-grade validation studies

- **Edge Deployment**

  Model distillation for on-campus and resource-constrained environments

Conclusion

Developed an end-to-end multimodal pipeline for educational content understanding with novel prompting strategies and evaluation metrics. Achieved significant improvements over all baselines. This practical tool supports adaptive learning systems and informed curriculum development for modern education.

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